
The Promises and Pitfalls of Spectral-Normalized Neural Gaussian Processes

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Abstract

Uncertainty estimation is important in high-stakes settings. Spectral-normalized Neural Gaussian Processes (SNGP) provide a distance-aware framework for uncertainty quantification. We investigate the impact of incorporating distance-based metric learning objectives, such as supervised contrastive learning (SupCon) and Multi-Similarity Loss, into the SNGP framework. Our empirical results show that these objectives provide only marginal improvements in accuracy and out-of-distribution detection while degrading calibration. We postulate that enforcing overly tight class clusters harms uncertainty estimation by promoting overconfident predictions near decision boundaries. To address this, we replace the standard Random Fourier Feature (RFF) approximation with Deterministic Uncertainty Estimation (DUE) and Gaussian Mixture Models (GMMs), which mitigate vanishing uncertainty and improve calibration. Finally, we demonstrate that leveraging self-supervised pretraining with DINOv2 yields a more semantically meaningful embedding space, achieving 97.49% accuracy, ECE 0.01, and 93.62% CIFAR-100 OOD AUPR — which amounts to more than 6% improvement over the best contrastive variant.

1 Introduction

Modern machine learning models are increasingly being deployed in high-stakes domains such as healthcare and autonomous systems, where uncertainty quantification is as important as predictive accuracy. Classical deep neural networks (DNNs) are powerful function approximators but tend to produce overconfident point estimates. Predictive scores derived from functions such as softmax are not reliable measures of uncertainty (17). Gaussian processes (GPs) are nonparametric Bayesian models that provide predictive distributions capturing epistemic uncertainty; however, they struggle to model complex, high-dimensional data. Therefore, it is natural to attempt to combine the uncertainty quantification of GPs with the modeling power of DNNs.

Spectral-normalized Neural Gaussian Processes (SNGP) (14) have emerged as a popular methodology for scalable uncertainty quantification with a DNN backbone. By combining spectral normalization with a Laplace-approximated GP output layer, SNGP produces distance-aware predictions that can accurately detect out-of-distribution samples. However, despite its promise, SNGP relies on two structural assumptions that to our knowledge have received limited scrutiny. First, it assumes that the input feature space is semantically meaningful and should be preserved in the hidden layers. Second, it relies on a Random Fourier Feature (RFF) approximation to the GP kernel, whose quality is sensitive to the dimensionality and structure of the feature space.

In this work, we present a systematic empirical study of these two limitations and evaluate several strategies to address them. First, to induce better label-sensitive geometry, we utilize a supervised contrastive learning objective (SupCon) (10) as well as Multi-Similarity (19). We find, however, that much of the observed empirical improvements are due to the associated data augmentation pipeline rather than the supervised contrastive objective function. Motivated by existing work demonstrating the usefulness of self-supervised learning (SSL) in constructing a more robust and semantically meaningful embedding space (7), we propose a SSL to SNGP finetuning methodology that demonstrates state of the art (SOTA) accuracy and out-of-distribution performance on image

data. Second, we examine approaches to circumvent the RFF GP approximation, including using inducing points methods and replacing the GP head entirely with a Gaussian Mixture Model (GMM).

Our contributions are as follows:

- We provide empirical analysis demonstrating that data augmentation under a contrastive learning paradigm boosts performance of classical SNGP training but supervised contrastive objectives do not aid the training process much beyond the signal provided by cross-entropy.
- We show that fine-tuning SNGP on top of a pretrained self-supervised model as a feature extractor leads to SOTA performance on image data.
- We evaluate both inducing point methods and GMMs as an alternative to RFF approximation in SNGP and study their interactions with contrastive learning objectives.

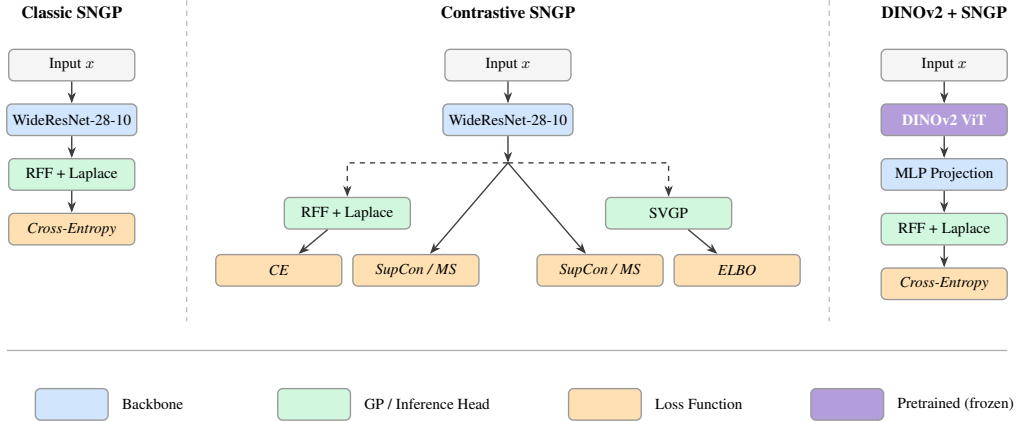


Figure 1: Overview of the three SNGP variants. The leftmost panel shows the classical SNGP training setup with RFF kernel and Laplace approximation combined with cross-entropy loss. The middle panel shows two different contrastive SNGP approaches we introduce. On the left, we keep the same RFF and Laplace approximation but introduce a supervised contrastive learning objective (SupCon / MS). On the right, we further replace the RFF + Laplace GP approximation with SVGP, a variational inference approach that uses the ELBO objective. The rightmost panel shows the self-supervised pretrained DINOv2 as the feature extractor for the classical SNGP setup.

2 Related Work

Distance-Aware Uncertainty Estimation A central challenge in deploying deep learning models is producing uncertainty estimates that are meaningful in the presence of distribution shift. Existing approaches for uncertainty estimation such as Monte Carlo Dropout (3), deep ensembles (12), and Bayesian neural networks (2) fail to explicitly encode distance awareness, which specifies that models should become increasingly more uncertain as test points move away from the training distribution to avoid overconfident uncertainty estimates. Liu et al. (14) introduces SNGP to induce this distance-awareness property in a deep neural network. The authors propose a GP output layer, which is naturally distance aware, combined with a bi-Lipschitz constraint on the hidden layer to ensure that the input geometry is faithfully preserved all the way to the GP layer. To make the GP scalable, SNGP uses a Random Fourier Feature (RFF) approximation for the kernel with a Laplace approximation for tractable posterior inference. SNGP represents an advance over prior uncertainty estimation methods for DNNs, but also has key limitations that we explore in more detail in Section 3.

Deterministic Uncertainty Estimation (DUE) Building on SNGP’s distance-preserving framework (14), DUE (18) enhances posterior quality while maintaining bi-Lipschitz constraints via spectral normalization. The critical differentiator is the Gaussian Process (GP) implementation: SNGP relies on RFF approximation, which treat the kernel as a linear model and can produce vanishing uncertainty estimates as training data grows. Conversely, DUE employs Stochastic Variational GPs (SVGP) (8). By optimizing inducing points via the Evidence Lower Bound (ELBO), DUE preserves non-parametric GP properties and ensures a better-calibrated uncertainty surface, offering a more rigorous probabilistic framework than the finite-feature approximation inherent in SNGP (20).

Supervised contrastive and metric learning losses. Khosla et al. (10) extend the self-supervised contrastive objective of SimCLR (1) to the fully labelled setting. Rather than treating only the other augmented view of the same image as a positive, Supervised Contrastive Learning (SupCon) treats all same-class examples in the batch as positives and all other-class examples as negatives, using a temperature-scaled softmax to simultaneously pull same-class embeddings together and push different-class embeddings apart. Wang et al. (19) propose Multi-Similarity Loss (MS-Loss), a metric learning objective that first mines pairs near the decision boundary and then applies self-adaptive, per-pair weights that grow with pair difficulty. MS-Loss was designed for image retrieval and has been shown to produce tighter, more discriminative embedding clusters than uniform-weight losses. Both losses are typically used as standalone training objectives for learning general-purpose visual representations. In this work we propose coupling them directly to the SNGP backbone to try and improve the geometry of the feature space seen by the GP head.

Self-supervised Learning Self-supervised learning provides a natural way to improve the geometry of visual representations. Existing work has proposed various unsupervised tasks to learn representations of a high-dimensional image space in a lower-dimensional semantic manifold that encode information about object identity, pose, lighting, texture, and scene structure (13; 9; 4). DINOv2, a large-scale pretrained self-supervised model, uses a student-teacher network pair to guide augmented views of the same image to a similar embedding which creates representations that preserve semantic information while discarding nuisance variation (16). These works focus on learning better representations of image data, which we leverage to create a semantically richer feature space as an input to SNGP.

We further analyze the effect of self-supervision on OOD detection and uncertainty estimation. Hendrycks et al. (7) found empirical evidence that self-supervision significantly improves robustness and performance across uncertainty estimation tasks. Our work extends this analysis by using SNGP as a final layer after a pretrained self-supervised feature extractor instead of a ResNet classifier head.

3 SNGP Formulation and Limitations

Backbone and Spectral Normalization We begin with a formal description of SNGP before identifying the two structural limitations that motivate this work. Let $h_\theta : \mathcal{X} \rightarrow \mathbb{R}^D$ denote the feature map parameterized by θ (in our case, a DNN). To encourage distance preservation in the feature space, SNGP applies spectral normalization to each weight matrix W in the backbone:

$$\tilde{W} = \frac{W}{\sigma(W)}$$

where $\sigma(W)$ is the largest singular value of W . This constrains the Lipschitz constant of each layer, encouraging the overall feature map to be approximately bi-Lipschitz:

$$L_1 \|x_1 - x_2\|_X \leq \|h_\theta(x_1) - h_\theta(x_2)\|_H \leq L_2 \|x_1 - x_2\|_X \quad (1)$$

for positive and bounded constants $0 < L_1 < 1 < L_2$ and a “suitable metric” for the input data manifold $\|\cdot\|_X$. Intuitively, this constraint prevents the feature map from arbitrarily collapsing or expanding distances that exist in the input space. We refer readers to the original paper (14) for more technical details of the SNGP formulation.

GP Layer and RFF Approximation SNGP replaces the typical dense output layer with a GP with an RBF kernel. Given N training samples $D = \{y_i, x_i\}_{i=1}^N$, the posterior distribution is computed as $p(g|D) \propto p(D|g)p(g)$ where $p(g)$ is the GP prior and $p(D|g)$ is the data likelihood for classification (cross-entropy loss).

The GP prior is approximated by deploying a low-rank decomposition of the kernel matrix $K \approx \Phi\Phi^\top$ via random features. The latent function values $\mathbf{g}$ are distributed according to a Multivariate Normal distribution:

$$\mathbf{g}_{N \times 1} \sim \text{MVN}(\mathbf{0}_{N \times 1}, \Phi\Phi_{N \times N}^\top)$$

Where the L -th hidden layer’s feature mapping $\Phi_{i,D_L} \in \mathbb{R}^{D_L \times 1}$ for an input h_i is defined as:

$$\Phi_{i,D_L} = \sqrt{\frac{2}{D_L}} \cos(W_L h_i + b_L) \quad (2)$$

where $W \in \mathbb{R}^{D \times d}$ is sampled from $\mathcal{N}(0, \sigma^2 I)$ and $b \sim \text{Uniform}(0, 2\pi)$. The GP predictive function is then expressed as a linear model $y(h) = \Phi(h)^\top \beta$ with a prior $\beta \sim \mathcal{N}(0, \sigma_\beta^2 I)$.

Limitation 1: Uninformative Input Geometry The bi-Lipschitz condition in Eq. 1 enforced by SNGP is only as useful as the input geometry it preserves. Spectral normalization constrains how the feature map transforms distances but does not determine what those distances mean semantically or even if the input geometry is beneficial to preserve.

During training, the cross-entropy objective induces label-sensitive structure in the hidden space but the resulting geometry is optimized for class discriminability on the training distribution, meaning that uncertainty over out-of-distribution data may be ill-formed. As a concrete illustration, we show a 2D representation in Figure 2 of the hidden feature space for CIFAR-10 and CIFAR-100 (11) of an SNGP model trained on CIFAR-10. Although cross-entropy has created clustered representations by label, the clusters are not tight and heavily overlap the CIFAR-100 representations. This overlap makes it more difficult to do effective out-of-distribution detection. We provide more detail about this visualization in Section 5.2.

Limitation 2: RFF Approximation Quality The RFF mapping in Eq.2 indicates that RFF relies on fixed basis functions sampled independently of the data. The fixed nature of the transformation mapping prevents the model from refining its underlying basis functions to match the actual geometric intricacies of the learned feature manifold. As demonstrated by van Amersfoort et al. (18), the RFF approximation in SNGP concentrates its uncertainty with large number of training samples, while being highly uncertain at low training samples. Consequently, the resulting variance calculations frequently diminish toward a state of absolute certainty in unexplored areas, failing to sustain an appropriate margin of error in the absence of local data points.

4 Supervised Contrastive Learning and Multi-Similarity Loss

Methods To address Limitation 1, we incorporate supervised metric learning objectives directly into SNGP training. Both losses operate on L_2 -normalised embeddings; since $\|z_i\| = 1$, their dot product equals cosine similarity $S_{ij} = z_i \cdot z_j \in [-1, 1]$. For anchor i , let $\mathcal{P}(i) = \{k \neq i : y_k = y_i\}$ be the positive set and $\mathcal{N}(i) = \{k : y_k \neq y_i\}$ the negative set.

Supervised Contrastive Loss (SupCon) (10). Using two independently augmented views per image (so $\mathcal{P}(i)$ includes both views of every same-class image in the batch), the loss for anchor i is:

$$\mathcal{L}_i^{\text{SC}} = -\frac{1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp(S_{ip}/\tau)}{\sum_{a \neq i} \exp(S_{ia}/\tau)}, \quad (3)$$

where $\tau=0.07$ is a temperature that sharpens the softmax, penalising negatives near the anchor more harshly. The total loss averages over all anchors. A key limitation is that every positive pair receives equal gradient weight regardless of how hard or easy it is.

Multi-Similarity Loss (MS-Loss) (19). MS-Loss first discards already-correct pairs via *hard-pair mining*, retaining only the informative subsets:

$$\mathcal{P}^*(i) = \{k \in \mathcal{P}(i) \mid S_{ik} < \max_{\mathcal{N}(i)} S_{ij} + \varepsilon\}, \quad \mathcal{N}^*(i) = \{k \in \mathcal{N}(i) \mid S_{ik} > \min_{\mathcal{P}^*(i)} S_{ij} - \varepsilon\},$$

with mining margin $\varepsilon=0.1$. It then weights each surviving pair adaptively:

$$\mathcal{L}_i^{\text{MS}} = \frac{1}{\alpha} \log\left(1 + \sum_{k \in \mathcal{P}^*(i)} e^{-\alpha(S_{ik}-\lambda)}\right) + \frac{1}{\beta} \log\left(1 + \sum_{k \in \mathcal{N}^*(i)} e^{\beta(S_{ik}-\lambda)}\right), \quad (4)$$

where $\lambda=0.5$ is the similarity threshold, $\alpha=2$ scales the positive penalty (large when same-class pairs are far apart), and $\beta=50$ scales the negative penalty (large when different-class pairs are dangerously close). The asymmetry $\beta \gg \alpha$ reflects that close negatives are more harmful than far positives. See Appendix A.4 for the full derivation.

In both cases the contrastive term is added to the cross-entropy loss. We also remove spectral normalization for the contrastive models, as the metric learning objective subsumes its geometric role (validated in Appendix A.2)

4.1 Model Variants and Training Protocols

Every experiment in this section is built on a Wide ResNet-28-10(WRN-28-10) (21) backbone feeding into a Random Feature GP (RFGP) classification head (14) with $D = 1024$ Random Fourier

Features. All models are trained on CIFAR-10 and evaluated on test accuracy, Expected Calibration Error (ECE), Negative Log-Likelihood (NLL), CIFAR-10-C corruption robustness (6), and OOD detection (AUPR) against SVHN and CIFAR-100. All results average over 5 random seeds and report mean $\pm$ std. Best model is chosen based on highest validation accuracy.

SNGP (baseline) uses minimal augmentation with RandomHorizontalFlip and RandomCrop. All other experiments use a stronger two-view pipeline, adapted from SimCLR (1), that applies random resized crops, strong colour jitter, and random grayscale conversion. These transformations are designed to reduce reliance on colour and low-level texture, encouraging the encoder to learn semantically invariant representations. We apply this pipeline twice independently to the same image, producing two differently augmented views of the same input. The experiments are as follows: (1) SNGP (Baseline), which uses WRN-28-10 with spectral normalisation on every convolutional layer and is trained with standard cross-entropy and the plain augmentation pipeline; (2) SNGP+Aug, which is identical to (1), but trained with the two-view augmentation described above; (3) SupCon+SNGP; and (4) MS-SNGP, which replaces SupCon with Multi-Similarity Loss.

Results Table 1 displays the results of the experiments. The largest single gain across all metrics comes from switching to the two-view augmentation pipeline (SNGP $\rightarrow$ SNGP+Aug): test accuracy improves by +1.75%, ECE drops from 0.05 to 0.02, NLL from 0.41 to 0.30, and CIFAR-10-C corruption accuracy jumps by +12.3 percentage points. OOD detection also improves markedly, with SVHN ds-AUPR rising from 95.71 to 97.32% and CIFAR-100 ds-AUPR from 83.18 to 86.78%. This result indicates that richer augmentation substantially regularises the GP’s feature space even without any explicit metric learning signal, likely because the stronger random crops and colour distortions force the encoder to build representations that are invariant to appearance variation within a class.

Both SupCon+SNGP and MS-SNGP improve on SNGP+Aug in accuracy (+0.76% and +0.47% respectively) and in OOD detection under the max-probability score (mp-AUPR: +4.5% on SVHN and +5.9% on CIFAR-100 for MS-SNGP). MS-SNGP achieves the best NLL (0.26), the best SVHN ds-AUPR (97.67%), and the best CIFAR-100 ds-AUPR (88.07%), with the pair-mining mechanism of MS-Loss producing tighter class clusters that sharpens the GP’s OOD discrimination. Despite the OOD and accuracy gains, both contrastive models exhibit worse calibration than SNGP+Aug: ECE increases from 0.02 to 0.04, and SupCon+SNGP’s CIFAR-10-C ECE reverts to 0.15, matching the unconstrained SNGP baseline. The tight embedding clusters enforced by the contrastive loss appear to make the GP slightly overconfident (see A.3).

In Section A.1 and A.2, we further examine the effect of the presence or absence of spectral norm, residual connections, or the dimension of the Random Fourier Feature approximation.

Table 1: Main results on CIFAR-10 and OOD benchmarks. Mean $\pm$ std over 5 random seeds. *ds-AUPR*: density-scaled AUPR (uses σ_c^2). *mp-AUPR*: max-probability AUPR (uses $1 - \max_c p_c^{MC}$). Best value in each column is **bold**.

Method	CIFAR-10			CIFAR-10-C		SVHN (OOD)		CIFAR-100 (OOD)	
	Acc (%)	ECE	NLL	Acc (%)	ECE	ds-AUPR (%)	mp-AUPR (%)	ds-AUPR (%)	mp-AUPR (%)
SNGP	91.92 $\pm$ 0.29	0.05 $\pm$ 0.03	0.41 $\pm$ 0.05	68.01 $\pm$ 1.44	0.15 $\pm$ 0.05	95.71 $\pm$ 1.05	92.48 $\pm$ 2.31	83.18 $\pm$ 0.50	75.91 $\pm$ 1.05
SNGP+Aug	93.67 $\pm$ 0.09	0.02 $\pm$ 0.01	0.30 $\pm$ 0.01	80.31 $\pm$ 0.49	0.08 $\pm$ 0.02	97.32 $\pm$ 0.30	91.74 $\pm$ 0.86	86.78 $\pm$ 0.19	78.67 $\pm$ 0.62
SupCon+SNGP	94.43 $\pm$ 0.12	0.04 $\pm$ 0.00	0.30 $\pm$ 0.01	81.22 $\pm$ 0.27	0.15 $\pm$ 0.00	96.74 $\pm$ 0.10	96.21 $\pm$ 0.19	85.06 $\pm$ 0.37	84.51 $\pm$ 0.38
MS-SNGP	94.14 $\pm$ 0.18	0.04 $\pm$ 0.00	0.26 $\pm$ 0.01	80.66 $\pm$ 0.47	0.12 $\pm$ 0.01	97.67 $\pm$ 0.46	96.79 $\pm$ 0.38	88.07 $\pm$ 0.22	86.65 $\pm$ 0.13
DUE	95.32 $\pm$ 0.19	0.03 $\pm$ 0.00	0.21 $\pm$ 0.01	74.41 $\pm$ 0.67	0.16 $\pm$ 0.01	93.39 $\pm$ 1.36	93.19 $\pm$ 1.39	84.36 $\pm$ 0.50	83.84 $\pm$ 0.46
DUE+Aug	95.11 $\pm$ 0.17	0.03 $\pm$ 0.00	0.21 $\pm$ 0.00	82.85 $\pm$ 0.53	0.10 $\pm$ 0.00	97.37 $\pm$ 0.43	97.00 $\pm$ 0.43	87.02 $\pm$ 0.49	86.37 $\pm$ 0.46
SupCon+DUE	95.12 $\pm$ 0.07	0.02 $\pm$ 0.00	0.20 $\pm$ 0.01	82.03 $\pm$ 0.09	0.10 $\pm$ 0.00	96.79 $\pm$ 0.50	96.42 $\pm$ 0.51	86.71 $\pm$ 0.48	85.94 $\pm$ 0.51
MS-DUE	94.48 $\pm$ 0.23	0.03 $\pm$ 0.00	0.21 $\pm$ 0.01	80.69 $\pm$ 0.63	0.09 $\pm$ 0.01	97.82 $\pm$ 0.46	97.48 $\pm$ 0.47	87.45 $\pm$ 0.38	86.67 $\pm$ 0.39
DINOv2 + SNGP	97.49 $\pm$ 0.05	0.01 $\pm$ 0.00	0.10 $\pm$ 0.01	82.44 $\pm$ 0.41	0.10 $\pm$ 0.02	99.34 $\pm$ 0.00	98.75 $\pm$ 0.00	94.85 $\pm$ 0.00	93.62 $\pm$ 0.00
DINOv2 + MLP	97.66 $\pm$ 0.06	0.01 $\pm$ 0.00	0.17 $\pm$ 0.01	82.94 $\pm$ 0.28	0.13 $\pm$ 0.00	99.45 $\pm$ 0.00	97.81 $\pm$ 0.00	96.03 $\pm$ 0.00	89.94 $\pm$ 0.00

5 Towards Better Geometry and Approximation in SNGP

5.1 Different Statistical Methods for Uncertainty Estimation

Given the calibration error results in Section 4 and the discussion in Section 3 regarding the limitations of the RFF approximation, it is evident that the supervised contrastive learning objective, which produces tighter intra-class clusters and increased inter-class separation, amplifies the tendency of RFF to produce overconfident, vanishing uncertainty estimates in out-of-distribution regions. This motivates a transition from SNGP to more expressive statistical frameworks.

5.1.1 Deterministic Uncertainty Estimation (DUE)

Formulation DUE (18) replaces the Random Fourier Feature (RFF) approximation used in SNGP with a Sparse Variational Gaussian Process (SVGP). The sparse GP introduces a set of M inducing variables $\mathbf{u}$ at inducing locations Z , which serve as a low-rank summary of the latent function. As illustrated in Figure 1, this allows DUE to retain the distance-aware structure of a GP while avoiding the fixed finite-feature approximation used by RFF. The model is trained using stochastic variational inference (SVI), which optimizes an evidence lower bound (ELBO). This objective balances the expected log-likelihood of the observed labels over minibatches with a KL-divergence penalty that keeps the variational posterior $q(\mathbf{u})$ close to the GP prior $p(\mathbf{u})$:

$$\mathcal{L} = \sum_{i=1}^N \mathbb{E}_{q(\mathbf{f}_i)} [\log p(y_i | \mathbf{f}_i)] - D_{KL}(q(\mathbf{u}) || p(\mathbf{u}))$$

Here, f_i denotes the latent GP function value associated with training input $\mathbf{x}_i$, which can be interpreted as the pre-softmax logit before being mapped to a class probability, and y_i denotes the corresponding observed class label for that training example. By marginalizing over the approximate posterior $q(\mathbf{u}) = \mathcal{N}(\mathbf{u} | \mathbf{m}, \mathbf{S})$, the model projects uncertainty from the inducing points to the rest of the feature space.

Results We follow the same experiment setup as 4.1 where we replace every implementation of RFF Approximation in SNGP with SVGP in DUE while maintaining same backbone. Table 1 second partition presents the CIFAR-10 and OOD benchmark metrics for the DUE framework. Comparative analysis against the SNGP results reveals that DUE yields consistent performance gains, with accuracy improvements ranging from +0.34% to +3.4% on CIFAR-10 and +0.03% to +6.4% on CIFAR-10-C. Furthermore, DUE demonstrates superior calibration, with the exception of the augmented variant, and a substantial reduction in NLL by 0.05 to 0.20. These findings validate our hypothesis that the RFF approximation in SNGP induces overconfident, vanishing uncertainty estimates that degrade both NLL and calibration.

A notable observation is that the margin of improvement provided by data augmentation is narrower in DUE than in SNGP. This suggests that the Sparse GP provides a more inherently robust uncertainty estimate, making the model less reliant on heavy augmentation to manage the distribution shifts observed in CIFAR-10-C. While the baseline DUE model achieves the highest accuracy on clean, in-distribution data, the augmented variant remains the most resilient to image corruptions.

The integration of metric learning objectives further refines these trade-offs: Supcon+DUE provides the best calibration, while MS-DUE variant achieves best OOD metrics, albeit with a slight reduction in predictive accuracy. Ultimately, these results indicate that the Sparse GP is more effective at leveraging the latent embedding space than the RFF head. However, the performance delta between standard and contrastive embeddings is less pronounced in DUE than in SNGP, suggesting that as the statistical approximation of the GP improves, the relative impact of the specific embedding strategy is partially moderated by the increased expressivity of the variational head.

5.1.2 Gaussian Mixture Model

We take a trained SNGP model and remove its prediction head, using the remaining encoder to extract embeddings. We then fit a Gaussian Mixture Model (GMM) to the training embeddings and their corresponding labels, thereby modeling the class-conditional densities in the learned embedding space.

Results First, the GMM-based approach underperforms the original SNGP model in terms of both NLL and accuracy. This suggests that a simple Gaussian mixture model may not be sufficiently expressive to capture the underlying structure of the embedding space learned by SNGP. The concrete results are reported in Appendix 5.

Second, Table 2 provides additional insight into the distributional properties of the learned embeddings. A GMM fitted on the CIFAR-10 *test* embeddings transfers well to the *training* embeddings, achieving near-perfect posterior accuracy. Similarly, a GMM fitted on the *training* embeddings performs well on the *test* embeddings, although its posterior accuracy is slightly lower than that of the test-fitted GMM. The larger difference appears in the fitted covariances: for every class, the train-fitted covariance has a smaller trace and a smaller largest eigenvalue than the corresponding test-fitted covariance. This suggests that the training embeddings occupy more compact class-conditional

regions, whereas the test embeddings are more dispersed. Under this interpretation, a GMM fitted only on training embeddings may underestimate the spread of the test embedding distribution, even though it captures the class centers well enough to maintain high classification accuracy.

Overall, these observations suggest that, although the embeddings learned with cross-entropy (CE) loss are highly separable across classes, their distributional geometry differs between the training and test data. The training embeddings form tighter class-conditional clusters, causing a GMM fitted on them to estimate narrower high-density regions than those observed in the test embeddings. Thus, even when class separability is high and posterior classification accuracy remains strong, the fitted density model can still be sensitive to differences in the spread of the embedding distribution. This highlights the importance of learning representations that are not only class-separable, but also distributionally stable between training and test data.

Setting	Value
<i>Posterior accuracy</i>	
GMM trained on CIFAR-10 training embeddings, evaluated on training set	0.9999
GMM trained on CIFAR-10 training embeddings, evaluated on test set	0.9286
GMM trained on CIFAR-10 test embeddings, evaluated on test set	0.9339
GMM trained on CIFAR-10 test embeddings, evaluated on training set	0.9999
<i>Covariance spread comparison: train-fitted GMM vs test-fitted GMM</i>	
Classes with smaller train-fitted covariance trace	10/10
Largest-eigenvalue ratio	0.449

Table 2: Posterior accuracy and covariance spread comparison for Gaussian Mixture Models (GMMs) fitted on CIFAR-10 training and test embeddings. Covariance ratios are computed class-wise as the train-fitted value divided by the test-fitted value; values below one indicate smaller spread for the train-fitted GMM.

5.2 Input Geometry Manipulation via Self-Supervised Learning

Methods and Rationale Given the results of supervised contrastive learning discussed in Section 4, understanding why data augmentation boosts the model’s understanding of both in-distribution and out-of-distribution data without the explicit use of labels may be beneficial. At a high level, data augmentation provides the model with different views of the same image, exposing it to different superficial transformations such as lighting, crop, and color jitter. The model can thus learn representations that are more invariant to superficial augmentations and learn a hidden geometry with semantic meaning beyond just the label. This type of representation learning can be viewed as a version of the self-supervised learning objective. Therefore, a natural extension to our work is to construct a new input geometry with self-supervised pretraining before passing the features to SNGP.

Setup We use DINOv2 (16) as a pretrained feature extractor and finetune a small SNGP model on top of the new features with CIFAR-10. To isolate the contribution of the SNGP head from the DINOv2 feature extractor, we evaluate two configurations: (1) DINOv2 with a standard MLP and softmax layer (DINOv2 + MLP) and (2) DINOv2 with an SNGP head (DINOv2 + SNGP). This ablation allows use to assess the benefit of SNGP on top of a semantically rich feature space.

Geometry Analysis Figure 2 shows UMAP projections of three hidden feature spaces: standard SNGP after CIFAR-10 training, DINOv2 + SNGP after CIFAR-10 finetuning, and DINOv2 + MLP after CIFAR-10 finetuning. As mentioned in Section 3, the standard SNGP feature space loosely clusters CIFAR-10 points (shown in color and as circles) with heavy overlap with CIFAR-100 points (shown in light gray and as triangles). Under DINOv2 with both SNGP and MLP heads, the class clusters are much tighter and the CIFAR-100 points have significantly less overlap. This visual representation indicates that DINOv2 is creating an input feature space that can distinguish between CIFAR-10 and CIFAR-100 without ever seeing CIFAR-100 labels. Further, we can see that there is undesired behavior in the DINOv2 + MLP feature space, such as the green class being separated into two different clusters which is not observed in the DINOv2 + SNGP feature space.

Results The third partition of Table 1 reports accuracy, ECE, NLL and OOD detection performance (ds-AUPR and mp-AUPR) across DINOv2 methods on CIFAR-10, CIFAR-10-C, SVHN, and CIFAR-100. DINOv2+SNGP and DINOv2+MLP together achieve state-of-the-art across nearly every metric,

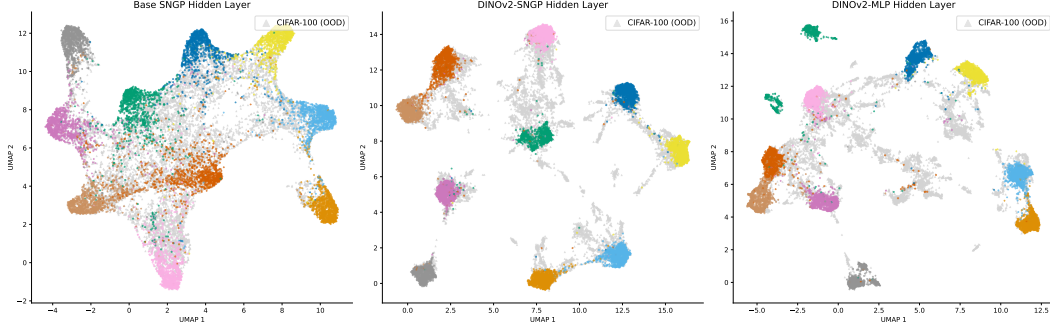


Figure 2: UMAP Visualization of Hidden Layers. The leftmost panel is the UMAP of the SNGP hidden layer (i.e. the vector passed to the GP head) trained on CIFAR-10 only, the center panel is the UMAP of the DINOv2 + SNGP hidden layer which was finetuned on CIFAR-10, and the rightmost panel is the UMAP of the DINOv2 + MLP hidden layer which was finetuned on CIFAR-10.

with DINOv2-based models improving substantially over all SNGP variants trained from scratch. Most notably, DINOv2+SNGP improves CIFAR-100 out of distribution detection by more than 6% compared to the best performing SNGP variant (mp-AUPR: 93.62, ds-AUPR: 94.85). Consistent with our hypothesis, these results indicate that the input geometry learned from the self-supervised feature extraction is beneficial for increased accuracy, calibration, and out-of-distribution detection.

Comparing DINOv2+SNGP directly against DINOv2+MLP highlights the tradeoff of the SNGP dual optimization structure. The MLP head achieves slightly higher in-distribution accuracy (97.66 vs 97.49) and marginally better performance on some OOD detection metrics, but the SNGP head achieves substantially better NLL and ECE on CIFAR-10 and CIFAR-10-C respectively. This result suggests that the GP head contributes to the calibration quality of the model that the MLP cannot provide despite the semantically rich feature space.

6 Limitations and Future Work

The Gaussian Mixture Model (GMM) experiments suggest that downstream density-based performance depends not only on intra-class compactness, but also on the stability and separation of class-conditional embedding distributions. Although cross-entropy encourages embeddings to be discriminative for classification, it does not explicitly constrain the geometry of the embedding space as a density model; in particular, it does not directly control class-conditional covariance, margins between class distributions, or the consistency of these quantities between training and test embeddings. As a result, the learned representation may achieve high classification accuracy while still producing class-conditional distributions whose spread is not fully aligned between training and test data. Future work could therefore investigate training objectives that impose more explicit geometric or distributional structure on the embedding space.

The contrastive losses employed in this work, particularly MS-Loss, introduce several hyperparameters (α , β , λ , ϵ) whose individual effects on embedding geometry and downstream GP calibration remain largely uncharacterized. A systematic sensitivity analysis is therefore needed to better understand how these parameters influence cluster compactness, inter-class margins, uncertainty estimates, and overall model performance.

Our experiments on self-supervised learning leveraged DINOv2, a model trained on a large and diverse dataset. Additional analyses and experiments are needed to ascertain whether similar performance gains can be achieved without external datasets or pretrained models.

7 Conclusion

In this work, we investigated two structural limitations of SNGP: uninformative input geometry and the quality of the RFF approximation. We find that data augmentation accounts for most of the gains attributed to contrastive learning, while contrastive objectives offer only marginal OOD improvements at the cost of calibration. Replacing the RFF head with a Sparse Variational GP (DUE) consistently improves calibration and NLL, confirming the insufficiency RFF approximation. Most interestingly, fine-tuning SNGP on DINOv2 features dominates all other design choices, proving that input geometry quality is the primary driver of uncertainty performance in distance-aware models.

References

- [1] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive self-supervised learning. In *Proceedings of the International Conference on Machine Learning*, pages 1597–1607, 2020.
- [2] Michael W. Dusenberry, Ghassen Jerfel, Yeming Wen, Yi-An Ma, Jasper Snoek, Katherine A. Heller, Balaji Lakshminarayanan, and Dustin Tran. Efficient and scalable bayesian neural nets with rank-1 factors. *CoRR*, abs/2005.07186, 2020. URL <https://arxiv.org/abs/2005.07186>.
- [3] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning, 2016. URL <https://arxiv.org/abs/1506.02142>.
- [4] Spyros Gidaris, Praveer Singh, and Nikos Komodakis. Unsupervised representation learning by predicting image rotations, 2018. URL <https://arxiv.org/abs/1803.07728>.
- [5] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.
- [6] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *arXiv preprint arXiv:1903.12261*, 2019.
- [7] Dan Hendrycks, Mantas Mazeika, Saurav Kadavath, and Dawn Song. Using self-supervised learning can improve model robustness and uncertainty. *CoRR*, abs/1906.12340, 2019. URL <http://arxiv.org/abs/1906.12340>.
- [8] James Hensman, Nicolo Fusi, and Neil D. Lawrence. Gaussian processes for big data, 2013. URL <https://arxiv.org/abs/1309.6835>.
- [9] Xu Ji, João F. Henriques, and Andrea Vedaldi. Invariant information clustering for unsupervised image classification and segmentation, 2019. URL <https://arxiv.org/abs/1807.06653>.
- [10] Prannay Khosla, Pang Tian, Chen Wang, Zhenyu Guo, Yonglong Tian, Brandon Smith, Phillip Isola, James Huggins, Ce Liu, Dilip Krishnan, and Yonglong Tian. Supervised contrastive learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 18661–18673, 2020.
- [11] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, April 2009. URL <https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf>.
- [12] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *arXiv preprint arXiv:1612.01474*, 2017. doi: 10.48550/arXiv.1612.01474.
- [13] Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Learning representations for automatic colorization, 2017. URL <https://arxiv.org/abs/1603.06668>.
- [14] Jeremiah Zhe Liu, Zi Lin, Shreyas Padhy, Dustin Tran, Tania Bedrax-Weiss, and Balaji Lakshminarayanan. Simple and principled uncertainty estimation with deterministic deep learning via distance awareness. *CoRR*, abs/2006.10108, 2020. URL <https://arxiv.org/abs/2006.10108>.
- [15] Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral normalization for generative adversarial networks. In *International Conference on Learning Representations*, 2018.
- [16] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision, 2024. URL <https://arxiv.org/abs/2304.07193>.

- [17] Tim Pearce, Alexandra Brintrup, and Jun Zhu. Understanding softmax confidence and uncertainty. *arXiv preprint arXiv:2106.04972*, 2021. doi: 10.48550/arXiv.2106.04972.
- [18] Joost van Amersfoort, Lewis Smith, Andrew Jesson, Oscar Key, and Yarin Gal. On feature collapse and deep kernel learning for single forward pass uncertainty. *arXiv preprint arXiv:2102.11409*, 2022. doi: 10.48550/arXiv.2102.11409.
- [19] Xun Wang, Xintao Han, Weilin Huang, Dengke Dong, and Matthew R. Scott. Multi-similarity loss with general pair weighting for deep metric learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5022–5030, 2019.
- [20] Andrew Gordon Wilson, Zhiting Hu, Ruslan Salakhutdinov, and Eric P. Xing. Stochastic variational deep kernel learning, 2016. URL <https://arxiv.org/abs/1611.00336>.
- [21] Sergey Zagoruyko and Nikos Komodakis. Wide residual networks. In *Proceedings of the British Machine Vision Conference*, 2016.

A Additional Details and Experiments

A.1 RFF Dimension Ablation

Table 3: Effect of the number of Random Fourier Features D on SNGP+Aug. All other hyperparameters are held fixed (WRN-28-10 backbone, spectral normalisation, strong two-view augmentation, cross-entropy loss, 200 epochs, 5 seeds).

Method	CIFAR-10			CIFAR-10-C		OOD AUPR (%)	
	Acc (%)	ECE	NLL	Acc (%)	ECE	SVHN (ds)	CIFAR-100 (ds)
SNGP+Aug ($D=512$)	93.59 $\pm$ 0.13	0.03 $\pm$ 0.00	0.26 $\pm$ 0.00	79.70 $\pm$ 0.37	0.12 $\pm$ 0.01	96.96 $\pm$ 0.47	86.21 $\pm$ 0.40
SNGP+Aug ($D=1024$)	93.67 $\pm$ 0.09	0.02 $\pm$ 0.01	0.30 $\pm$ 0.01	80.31 $\pm$ 0.49	0.08 $\pm$ 0.02	97.32 $\pm$ 0.30	86.78 $\pm$ 0.19
SNGP+Aug ($D=4096$)	93.79 $\pm$ 0.21	0.03 $\pm$ 0.00	0.26 $\pm$ 0.01	80.07 $\pm$ 0.36 [†]	0.12 $\pm$ 0.00 [†]	97.46 $\pm$ 0.27	87.00 $\pm$ 0.43

The number of Random Fourier Features D controls the quality of the kernel approximation in the RFGP head. Recall from Eq. (7) that the posterior variance is computed via a rank- D precision matrix $\Lambda_c \in \mathbb{R}^{D \times D}$; larger D gives a richer approximation to the exact GP covariance, at the cost of quadratic memory in D and a slower inverse at test time.

The results in Table 3 show that **performance gradually improves as D is increased across the range [512, 4096]**. We picked $D=1024$ as our default in the other experiments as it captures nearly all the OOD benefit of $D=4096$ at one quarter of the memory cost. The marginal gains from further increasing D do not justify the added computational overhead for CIFAR-10-scale experiments.

A.2 Ablation: Residual Connections and Spectral Normalisation with Contrastive Loss

Table 4: Ablation study: spectral normalisation, residual connections, and hybrid contrastive objectives. All variants use WRN-28-10 with the strong two-view augmentation pipeline and MS-Loss (plus cross-entropy), trained for 200 epochs over 5 seeds (4 for MS-SNGP NoSkip due to one time-limited job). Only the architectural choices listed differ from the MS-SNGP baseline. *ds-AUPR*: density-scaled AUPR. *mp-AUPR*: max-probability AUPR.

Method	CIFAR-10			CIFAR-10-C		SVHN (OOD)		CIFAR-100 (OOD)	
	Acc (%)	ECE	NLL	Acc (%)	ECE	ds-AUPR (%)	mp-AUPR (%)	ds-AUPR (%)	mp-AUPR (%)
MS-SNGP (baseline)	94.14 $\pm$ 0.18	0.04 $\pm$ 0.00	0.26 $\pm$ 0.01	80.66 $\pm$ 0.47	0.12 $\pm$ 0.01	97.67 $\pm$ 0.46	96.79 $\pm$ 0.38	88.07 $\pm$ 0.22	86.65 $\pm$ 0.13
MS-SNGP+SN	94.05 $\pm$ 0.19	0.04 $\pm$ 0.00	0.28 $\pm$ 0.01	80.58 $\pm$ 0.66	0.13 $\pm$ 0.01	97.60 $\pm$ 0.27	96.49 $\pm$ 0.30	87.58 $\pm$ 0.50	86.05 $\pm$ 0.48
MS-SNGP NoSkip	92.08 $\pm$ 0.39	0.05 $\pm$ 0.00	0.37 $\pm$ 0.01	76.56 $\pm$ 0.57	0.16 $\pm$ 0.00	96.43 $\pm$ 0.41	95.43 $\pm$ 0.43	84.69 $\pm$ 0.56	82.46 $\pm$ 0.61
Hybrid SNGP	94.14 $\pm$ 0.12	0.05 $\pm$ 0.00	0.30 $\pm$ 0.01	80.63 $\pm$ 0.20	0.15 $\pm$ 0.01	97.97 $\pm$ 0.29	96.80 $\pm$ 0.35	86.30 $\pm$ 0.26	85.37 $\pm$ 0.63

Spectral normalisation is redundant with contrastive loss. MS-SNGP+SN re-introduces spectral normalisation (per-layer Lipschitz coefficient ≤ 1 (15)) on top of the MS-Loss training. Table 4 shows that this has negligible effect: accuracy, ECE, NLL, and all OOD metrics are within one standard deviation of the baseline MS-SNGP on every column. The reason is probably as follows: SN’s purpose is to constrain each layer’s Lipschitz constant so that inputs far apart in pixel space

remain far apart in embedding space, providing the GP’s uncertainty estimate with a meaningful notion of distance. However, the contrastive loss already optimises pairwise distances in embedding space at every training step, directly enforcing the cluster geometry. Once the encoder has been trained with MS-Loss, the additional Lipschitz constraint imposed by SN neither tightens the clusters further nor improves OOD sensitivity. So, confirming our intuition, removing SN also simplifies the model and removes a source of potential over-constraint during early training.

Residual connections are essential. Removing skip connections (MS-SNGP NoSkip) causes a consistent and substantial degradation across all metrics: test accuracy drops by -2.06% , NLL worsens from 0.26 to 0.37, CIFAR-10-C accuracy falls by -4.1 percentage points, and SVHN / CIFAR-100 OOD AUPR drop by -1.2% and -3.4% respectively. Calibration also degrades (ECE 0.05, CIFAR-10-C ECE 0.16), and the qualitative analysis in Section A.3 reveals that MS NoSkip undergoes a near-complete *uncertainty collapse* — $\sigma_c^2 \approx 0$ for all inputs including OOD, rendering the GP’s epistemic signal useless. The root cause is that skip connections serve two complementary roles in the contrastive setting. First, they provide low-loss gradient paths (5) that allow the contrastive signal to propagate back into the early convolutional layers; without them, only the later layers receive meaningful gradient, and the early features converge to a fixed point independent of the input class. Second, the identity shortcut preserves input-space distances across layers, maintaining the Lipschitz structure that gives the GP posterior variance its geometric meaning.

Combining SupCon and MS-Loss does not improve over either alone. In our Hybrid model, the training objective combines both contrastive losses with cross-entropy:

$$\mathcal{L} = w_{\text{SC}} \mathcal{L}_{\text{SupCon}} + w_{\text{MS}} \mathcal{L}_{\text{MS}} + \mathcal{L}_{\text{CE}}. \quad (5)$$

It matches MS-SNGP on accuracy and achieves a marginally higher SVHN AUPR (97.97% vs 97.67%), but underperforms MS-SNGP on CIFAR-100 OOD detection (86.30% vs 88.07%). It also inherits SupCon’s worse calibration: CIFAR-10-C ECE rises to 0.15 (matching the unconstrained SNGP baseline). The interference is interpretable: SupCon’s uniform gradient over all pairs partially cancels MS-Loss’s selective, difficulty-weighted updates, preventing the hard-pair refinement that makes MS-SNGP the stronger OOD detector. Adding a second contrastive objective thus dilutes the training signal.

A.3 Qualitative Uncertainty Analysis Experiments

Standard evaluation metrics such as accuracy, ECE, and OOD AUPR aggregate over thousands of examples and can obscure how a model forms its uncertainty estimates on individual inputs. So, we designed a structured 20-image probe that spans four regimes: images drawn from the test set of the training distribution, real-world photographs that are visually clear but not identical to training data, real-world photographs that are ambiguous between two CIFAR-10 classes, and images that lie completely outside the CIFAR-10 label space.

A.3.1 Measured Quantities

All models share the same inference structure. Let $\phi(x) \in \mathbb{R}^D$ be the RFF embedding of input x , where D is the number of inducing features (1024 by default across the experiments discussed here). The GP mean (pre-softmax logit) for class c is

$$f_c(x) = \mathbf{w}_c^\top \phi(x), \quad (6)$$

and the corresponding GP posterior variance—accumulated during training via the precision matrix $\mathbf{\Lambda}_c$ —is

$$\sigma_c^2(x) \approx \frac{1}{\lambda} - \phi(x)^\top \left(\mathbf{\Lambda}_c + \frac{\mathbf{I}}{\lambda} \right)^{-1} \phi(x), \quad (7)$$

where λ is the ridge penalty. Intuitively, $\sigma_c^2(x)$ is the fraction of the GP prior variance that was *not* explained by training data: it is small when $\phi(x)$ lies in a direction that $\mathbf{\Lambda}_c$ has accumulated mass along (i.e. training examples have been seen nearby), and large when $\phi(x)$ is novel. This makes it a direct measure of model uncertainty.

We report two uncertainty-aware predictive probabilities. The deterministic Laplace approximation applies a closed-form temperature correction:

$$p_c^{\text{det}}(x) = \text{softmax}\left(\frac{f_c(x)}{\sqrt{1 + \frac{\pi}{8} \sigma_c^2(x)}}\right), \quad (8)$$

which scales each logit by the standard deviation of the implied Gaussian, reducing confidence when epistemic uncertainty is high. The Monte Carlo Laplace estimate averages over $S = 50$ samples from the approximate posterior:

$$p_c^{\text{MC}}(x) = \frac{1}{S} \sum_{s=1}^S \text{softmax}\left(\tilde{\mathbf{f}}^{(s)}\right)_c, \quad \tilde{\mathbf{f}}^{(s)} \sim \mathcal{N}(\mathbf{f}(x), \text{diag}(\sigma^2(x))). \quad (9)$$

All results reported below use p_c^{MC} as the primary predictive probability; σ_c^2 refers to the per-class GP posterior variance of the predicted (argmax) class. As for the OOD experiments, those of the top class instead of the true class are reported.

A.3.2 Experiment Setup

The 20 images are divided into four groups of five:

1. **CIFAR-10 test images.** Five images sampled from the held-out CIFAR-10 test set (one per class: airplane, automobile, cat, horse, ship) using a fixed random seed.
2. **Clear real-world photographs.** Five high-quality photographs whose subject belongs to a single CIFAR-10 class (German shepherd $\rightarrow$ dog; Mercedes Actros $\rightarrow$ truck; container ship $\rightarrow$ ship; European robin $\rightarrow$ bird; fallow deer $\rightarrow$ deer). These test whether the models generalise beyond the low-resolution CIFAR-10 rendering.
3. **Ambiguous real-world photographs.** Five photographs whose subject could reasonably be assigned to one of two CIFAR-10 classes: Shiba Inu (dog / cat), Suzuki Carry kei-truck (truck / automobile), hovercraft (ship / automobile), osprey in flight (bird / airplane), and sika deer (deer / horse). These test inter-class uncertainty estimation.
4. **Out-of-distribution (OOD) images.** Five images whose semantic content has no counterpart in CIFAR-10: a moose, the Hubble Space Telescope, a garden snail, a brown bear, and an oak tree. These subjects are visually related to CIFAR-10 classes (moose $\approx$ deer, satellite $\approx$ airplane, snail $\approx$ frog, bear $\approx$ dog, tree $\approx$ none) but do not belong to them.

All images are rescaled to 32×32 pixels to match the model input resolution.

A.3.3 CIFAR-10 Test Images: Identifying Collapsed Models

runs.summary["tables/CIFAR10_Test"]

Filter

	Image	True label	MS-SNGP pred	MS-SNGP σ^2	MS-SNGP p_mc	MS-SNGP+SN pred	MS-SNGP+SN σ^2	MS-SNGP+SN p_mc	MS-SNGP+SN pred	MS-SNGP+SN σ^2	MS-SNGP+SN p_mc	MS-SNGP+SN pred	MS-SNGP+SN σ^2	MS-SNGP+SN p_mc
1		airplane	✓ airplane airplane 0.91 / automobile 0.03 / frog 0.02	2.827	0.912	✓ airplane airplane 0.91 / automobile 0.03 / ship 0.02	2.601	0.912	✓ airplane airplane 0.99 / ship 0.00 / bird 0.00	0.004	0.986	✓ airplane airplane 0.96 / automobile 0.01 /	1.679	0.964
2		automobile	✓ automobile automobile 0.97 / deer 0.02 / bird 0.00	3.509	0.97	✓ automobile automobile 0.97 / horse 0.03 / deer 0.00	3.062	0.966	✓ automobile automobile 0.99 / truck 0.00 / ship 0.00	0.005	0.995	✓ automobile automobile 1.00 / truck 0.00 /	0.279	1
3		cat	✓ cat cat 0.47 / truck 0.22 / frog 0.09	1.949	0.473	✓ cat cat 0.97 / ship 0.02 / truck 0.00	1.665	0.971	✓ cat cat 0.93 / dog 0.04 / automobile 0.01	0.003	0.93	✓ cat cat 1.00 / dog 0.00 / frog 0.00	0.388	0.998
4		horse	✓ horse horse 0.90 / ship 0.06 / bird 0.01	3.41	0.903	✓ horse horse 0.96 / automobile 0.02 / frog 0.01	3.653	0.961	✓ horse horse 0.98 / dog 0.01 / deer 0.00	0.002	0.98	✓ horse horse 1.00 / automobile 0.00 / dog 0.00	0.419	1
5		ship	✓ ship ship 0.94 / automobile 0.01 / frog 0.01	6.553	0.945	✓ ship ship 0.96 / bird 0.01 / truck 0.01	5.967	0.958	✓ ship ship 0.99 / airplane 0.00 / automobile 0.00	0.003	0.993	✓ ship ship 1.00 / airplane 0.00 / frog 0.00	0.181	1

Figure 3: Per-model predictions and uncertainty on genuine CIFAR-10 test images. Five held-out test images (airplane, automobile, cat, horse, ship) are evaluated on all seven model variants. Each cell reports the top-3 class probabilities under the Monte Carlo Laplace approximation (p_c^{MC}), the GP posterior variance σ_c^2 , and the hard prediction.

runs.summary["tables/CIFAR10_Test"]

Filter

	Image	True label	MS-SNGP+SN pred	MS-SNGP+SN σ^2	MS-SNGP+SN p_mc	MS NoSkip pred	MS NoSkip σ^2	MS NoSkip p_mc	Hybrid pred	Hybrid σ^2	Hybrid p_mc
1		airplane	✗ ship ship 0.18 / horse 0.16 / truck 0.11	9.281	0.1	✓ airplane airplane 1.00 / ship 0.00 / bird 0.00	0.153	0.999	✗ automobile automobile 0.19 / bird 0.16 / ship 0.13	1.643	0.114
2		automobile	✗ horse horse 0.15 / automobile 0.14 / bird 0.13	11.822	0.141	✓ automobile automobile 1.00 / truck 0.00 / ship 0.00	0.119	1	✗ truck truck 0.17 / airplane 0.14 / ship 0.14	4.798	0.113
3		cat	✗ horse horse 0.28 / ship 0.14 / automobile 0.12	4.546	0.061	✓ cat cat 1.00 / dog 0.00 / automobile 0.00	0.085	0.998	✗ horse horse 0.15 / automobile 0.14 / truck 0.14	0.816	0.05
4		horse	✗ airplane airplane 0.20 / ship 0.16 / automobile 0.14	10.12	0.084	✓ horse horse 1.00 / deer 0.00 / dog 0.00	0.096	1	✗ automobile automobile 0.17 / truck 0.14 / deer 0.13	2.702	0.112
5		ship	✗ airplane airplane 0.14 / frog 0.13 / bird 0.12	10.923	0.053	✓ ship ship 1.00 / airplane 0.00 / truck 0.00	0.064	1	✗ automobile automobile 0.19 / frog 0.14 / deer 0.13	4.067	0.086

Figure 4: Per-model predictions and uncertainty on genuine CIFAR-10 test images: failed models. Five held-out test images (airplane, automobile, cat, horse, ship) are evaluated on all seven model variants. Each cell reports the top-3 class probabilities under the Monte Carlo Laplace approximation (p_c^{MC}), the GP posterior variance σ_c^2 , and the hard prediction. Models whose posterior collapsed to near-uniform outputs (MS-SNGP+SN, Hybrid SNGP) or whose variance collapsed to near-zero (MS NoSkip) are immediately identifiable here and excluded from the subsequent analyses.

Figure 3 and 4 reveals immediate, irrecoverable pathologies in three of the seven variants.

MS-SNGP+SN and Hybrid SNGP: posterior collapse. Both models produce near-uniform predictive distributions across all five test images, achieving 0/5 correct classifications. Their GP logits $f_c(x)$ are extremely small in magnitude and nearly class-indiscernible. The root cause is a failure of the GP readout layer to learn a meaningful posterior: the spectral normalisation constraint

in MS-SNGP+SN, applied on top of the contrastive training regime, over-constrains the backbone’s Lipschitz constant to the point that the RFF projection $\phi(x)$ loses discriminative information before reaching the GP head. The Hybrid SNGP applies a *joint* cross-entropy + supcon + MS; the three objectives might be interfering in a destructive way, again preventing a coherent posterior from forming. Both models are excluded from all subsequent analyses.

MS NoSkip: uncertainty collapse. The no-skip-connection variant achieves 5/5 correct classifications with high p^{MC} on the predicted class, superficially resembling a well-trained model. However, $\sigma_c^2(x)$ is near-zero (0.05) for every image regardless of its content. The collapse occurs because removing skip connections eliminates the identity paths that ordinarily allow gradient signal from the loss to propagate back into the early layers. Without these paths, early-layer features converge to a single fixed point independent of the input, so every $\phi(x)$ lands in the same narrow region of the RFF space. The precision matrix $\mathbf{\Lambda}_c$ accumulates mass uniformly in that region, and Equation (7) evaluates to nearly zero for all inputs—both in-distribution and OOD. Since σ_c^2 carries no information, the model is useless as an uncertainty estimator despite its high accuracy. MS NoSkip is likewise excluded from further analysis.

Surviving models. The remaining four variants—SNGP, SNGP+Aug, SupCon+SNGP, and MS-SNGP—all achieve 5/5 correct classifications on the CIFAR-10 test images and exhibit σ_c^2 values that are small but non-zero, indicating healthy GP posteriors.

A.3.4 Real-World Photographs: Inter-Class Uncertainty and Overconfidence

runs.summary["tables/Clear_easy"]

Image	True label	SNGP pred	SNGP σ^2	SNGP p_n	SNGP+Aug pred	SNGP+Aug σ^2	SNGP+Aug p_m	SupCon+SNGP pred	SupCon+SNGP σ^2	SupCon+SNGP p_mc	MS-SNGP pred	MS-SNGP σ^2	MS-SNGP p_mc
1 	dog	✗ horse horse 0.66 / ship 0.09 / automobil 0.09	19.628	0.039	✗ horse horse 0.93 / automobile 0.05 / truck 0.01	11.046	0.001	✗ horse horse 0.81 / airplane 0.11 / cat 0.03	0.007	0.018	✗ horse horse 0.98 / ship 0.00 / airplane 0.00	3.936	0.002
2 	truck	✓ truck truck 0.92 / bird 0.03 / deer 0.02	3.463	0.918	✓ truck truck 1.00 / frog 0.00 / ship 0.00	2.958	0.997	✓ truck truck 0.99 / automobile 0.00 / ship 0.00	0.004	0.992	✓ truck truck 1.00 / airplane 0.00 / automobil 0.00	0.678	0.999
3 	ship	✓ ship ship 0.87 / truck 0.04 / horse 0.04	8.595	0.865	✓ ship ship 0.94 / frog 0.02 / deer 0.01	4.624	0.939	✗ airplane airplane 0.81 / ship 0.17 / bird 0.01	0.005	0.172	✓ ship ship 0.99 / airplane 0.00 / automobil 0.00	1.02	0.994
4 	bird	✓ bird bird 0.95 / automobil 0.02 / truck 0.02	3.241	0.946	✓ bird bird 0.97 / truck 0.01 / airplane 0.00	3.225	0.972	✓ bird bird 0.94 / airplane 0.02 / cat 0.01	0.003	0.942	✓ bird bird 1.00 / dog 0.00 / deer 0.00	0.309	0.999
5 	deer	✓ deer deer 0.89 / automobil 0.06 / ship 0.03	2.744	0.89	✓ deer deer 0.97 / truck 0.03 / automobile 0.00	2.458	0.967	✓ deer deer 0.86 / cat 0.08 / bird 0.02	0.003	0.86	✓ deer deer 1.00 / bird 0.00 / horse 0.00	0.205	1

Figure 5: Predictions and uncertainty on unambiguous real-world photographs. Five real photographs whose true label maps cleanly to a single CIFAR-10 class are evaluated on the four models (SNGP, SNGP+Aug, SupCon+SNGP, MS-SNGP). The GP posterior variance σ_c^2 remains comparable to that observed on genuine CIFAR-10 test images, confirming that all four models treat high-quality, canonical real-world photographs as in-distribution.

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Filter

	Image	True label	SNGP pred	SNGP σ^2	SNGP p_...	SNGP+Aug pred	SNGP+Aug σ^2	SNGP+Aug p_mc	SupCon+SNGP pred	SupCon+SNGP σ^2	SupCon+MS-SNGP pred	MS-SNGP σ^2	MS-SNGP p_mc
1		dog	✓ dog dog 0.84 / ship 0.07 / frog 0.02	3.169	0.839	✓ dog dog 0.95 / truck 0.03 / automobile 0.02	2.555	0.945	✓ dog dog 0.97 / cat 0.01 / horse 0.01	0.002	0.975	✓ dog dog 1.00 / cat 0.00 / ship 0.00	0.192
2		truck	✗ automobile automobile 0.44 / truck 0.19 / ship 0.07	1.752	0.194	✗ automobile automobile 0.96 / ship 0.03 / dog 0.01	3.458	0	✗ automobile automobile 0.87 / truck 0.12 / ship 0.00	0.003	0.121	✗ automobil automobil 1.00 / truck 0.00 /	0.471
3		ship	✓ ship ship 0.92 / frog 0.04 / bird 0.02	2.632	0.922	✓ ship ship 0.97 / frog 0.02 / dog 0.01	1.825	0.97	✓ ship ship 0.98 / airplane 0.01 / truck 0.01	0.003	0.983	✓ ship ship 1.00 / airplane 0.00 / truck 0.00	0.19
4		bird	✓ bird bird 0.76 / ship 0.10 / automobile 0.04	2.318	0.763	✓ bird bird 0.97 / truck 0.02 / frog 0.00	3.033	0.968	✓ bird bird 0.95 / airplane 0.03 / cat 0.00	0.002	0.95	✓ bird bird 1.00 / frog 0.00 / deer 0.00	0.185
5		deer	✓ deer deer 0.90 / automobile 0.05 / truck 0.03	4.171	0.897	✓ deer deer 0.94 / ship 0.03 / horse 0.01	5.029	0.936	✓ deer deer 0.98 / dog 0.00 / cat 0.00	0.002	0.984	✓ deer deer 1.00 / bird 0.00 / cat 0.00	0.295

Figure 6: Predictions and uncertainty on visually ambiguous real-world photographs. Five images that straddle two CIFAR-10 classes are evaluated: Shiba Inu (dog / cat), Suzuki Carry kei-truck (truck / automobile), hovercraft (ship / automobile), osprey in flight (bird / airplane), and sika deer (deer / horse). Differences in *inter-class* uncertainty between models are most visible here: contrastively trained models (SupCon+SNGP, MS-SNGP) tend toward sharper, lower-entropy predictions despite genuine visual ambiguity, while the plain SNGP shows softer probability mass across the confusable classes.

Figures 5 and 6 show the four surviving models on real-world photographs.

All four models treat high-quality photographs as in-distribution. For both the clear and ambiguous sets, the GP posterior variance σ_c^2 remains at magnitudes comparable to those observed on genuine CIFAR-10 test images ($\sigma_c^2 \sim 0.1$ – 0.4 for SNGP and SNGP+Aug, $\sigma_c^2 \sim 0.02$ – 0.15 for SupCon+SNGP and MS-SNGP). This is expected: high-quality colour photographs of CIFAR-10-class objects, once downsampled to 32×32 , produce RFF embeddings $\phi(x)$ that fall within the support accumulated in Λ_c during training. The models correctly report low epistemic uncertainty.

Misclassification of the clear Dog. The German shepherd (clear, dog) is misclassified as *horse* by all four models, likely because the image has other objects in it too. Importantly, all four models report this misclassification with moderate-to-high confidence ($p_{\text{horse}}^{\text{MC}} > 0.6$), yet with σ_c^2 values that are no higher than those of correctly classified images. This illustrates the fundamental limit of epistemic uncertainty: the model does not know that it does not know as it wasn’t trained on such images before.

Inter-class uncertainty: Limitation of MS Loss The Suzuki Carry kei-truck is labelled as *truck* but is visually nearly identical to a compact automobile. SNGP assigns $p_{\text{truck}}^{\text{MC}} \approx 0.25$ and $p_{\text{automobile}}^{\text{MC}} \approx 0.47$ —a genuinely diffuse distribution that reflects the visual ambiguity. MS-SNGP, by contrast, assigns $p_{\text{automobile}}^{\text{MC}} \approx 1.00$ with near-zero mass on any other class. The multi-similarity contrastive loss pushes embeddings of same-class examples tightly together and pushes embeddings of different-class examples apart. After training, the boundary between the *truck* and *automobile* embedding clusters is sharp, and the Suzuki Carry’s embedding happens to land on the automobile side. Because the GP precision matrix Λ_c has dense support on both sides of that boundary, σ_c^2 is low everywhere near it, and the model commits confidently. The contrastive loss thus trades *intra-class* uncertainty (the model is never confused within a class) for *inter-class* uncertainty (the model cannot express doubt about which class to assign). For applications where confident misclassification is more dangerous than abstention, this is a clear limitation.

runs.summary["tables/OOD_not_CIFAR10"]

Filter

	Image	True label	SNGP pred	SNGP σ^2	SNGP+Aug pred	SNGP+Aug σ^2	SupCon+SNGP	SupCon+SNGP σ^2	MS-SNGP pred	MS-SNGP σ^2
1		OOD	deer deer 0.85 / airplane 0.06 / automobile 0.05	63.361	deer deer 0.85 / horse 0.10 / automobile 0.01	15.067	horse horse 0.61 / deer 0.33 / dog 0.04	0.054	deer deer 0.73 / horse 0.13 / automobile 0.05	15.204
2		OOD	airplane airplane 0.96 / automobile 0.02 / truck 0.01	48.132	airplane airplane 0.49 / bird 0.35 / automobile 0.04	26.936	airplane airplane 0.65 / bird 0.31 / ship	0.02	bird bird 0.41 / airplane 0.28 / automobile 0.12	16.254
3		OOD	cat cat 0.73 / automobile 0.07 / truck 0.07	38.815	truck truck 0.82 / automobile 0.08 / ship 0.04	18.434	frog frog 0.39 / cat 0.26 / deer 0.12	0.012	ship ship 0.94 / frog 0.02 / truck 0.01	10.001
4		OOD	deer deer 0.91 / automobile 0.04 / frog 0.02	63.807	dog dog 0.43 / deer 0.37 / automobile 0.07	47.359	deer deer 0.97 / dog 0.01 / horse 0.01	0.073	deer deer 0.99 / horse 0.00 / airplane 0.00	6.064
5		OOD	ship ship 0.40 / truck 0.17 / automobile 0.11	54.66	truck truck 0.54 / ship 0.30 / frog 0.04	23.825	ship ship 0.84 / truck 0.07 / airplane 0.07	0.017	ship ship 0.99 / truck 0.00 / airplane 0.00	6.729

Figure 7: Predictions and epistemic uncertainty on out-of-distribution images. Five images whose semantic content has no CIFAR-10 class are evaluated. The GP posterior variance σ_c^2 rises substantially for all four healthy models, indicating that the RFF-based covariance correctly detects the distributional shift. The magnitude of the spike, and its consistency across the five OOD images, varies between models.

A.3.5 Out-of-Distribution Images: Epistemic Uncertainty Spikes

Figure 7 shows the OOD results.

All four models detect distributional shift via σ_c^2 . For the five OOD images, every surviving model exhibits a marked increase in σ_c^2 relative to its in-distribution baseline, confirming that the RFF covariance correctly identifies novel embeddings. To quantify this, we define the *OOD spike ratio* as the mean σ_c^2 over the five OOD images divided by the mean σ_c^2 over the ten in-distribution images (CIFAR-10 test + clear real-world):

$$\rho = \frac{\overline{\sigma_c^2} [\text{OOD}]}{\overline{\sigma_c^2} [\text{in-distribution}]}.$$
 (10)

The observed spike ratios are approximately $\rho_{\text{SNGP}} \approx 10$, $\rho_{\text{SNGP+Aug}} \approx 6$, $\rho_{\text{MS-SNGP}} \approx 22$, and $\rho_{\text{SupCon+SNGP}} \approx 13$. MS-SNGP achieves the highest absolute OOD-to-in-distribution contrast, which is consistent with its leading AUPR on the quantitative OOD benchmarks (SVHN and CIFAR-100).

A.3.6 Future Work: Uncertainty Along an OOD Trajectory

A natural extension of this would track $\sigma_c^2(x_\alpha)$ along a continuous interpolation between an in-distribution image and its fully corrupted counterpart:

$$x_\alpha = (1 - \alpha) x_{\text{in}} + \alpha \eta, \quad \alpha \in [0, 1],$$
 (11)

where η is structured OOD image. One would expect σ_c^2 to increase monotonically with α for a well-calibrated model, and the rate of increase should reflect how quickly the embedding $\phi(x_\alpha)$ moves out of the training support. Contrastively trained models, with their tight cluster geometry, might show a sharper, later transition—remaining confident almost until α is large—whereas the plain SNGP might degrade more gracefully.

A.4 Explanation of Contrastive Losses

The GP’s uncertainty is inherently distance-based, where the variance $\sigma_c^2(x)$ (Eq. 7) remains small only when $\phi(x)$ aligns with directions where the precision matrix Λ_c has accumulated mass. For this signal to be semantically meaningful, both losses actively shape the embedding space by mapping each image to an L_2 -normalized vector $z_i \in \mathbb{R}^D$ on the unit hypersphere $\mathbb{S}^{D-1}$. By utilizing Metric Learning to ensure that same-class embeddings cluster tightly while different-class embeddings remain well-separated, the GP head receives a structured representation where geometric distance directly corresponds to class identity.

Since all vectors are unit-norm, their dot product equals their cosine similarity:

$$S_{ij} = z_i \cdot z_j \in [-1, 1]. \quad (12)$$

For a batch of N embeddings with integer labels $y_1, \dots, y_N$, fix anchor i and define:

- $\mathcal{P}(i)$ — **positive set**: all $k \neq i$ with $y_k = y_i$.
- $\mathcal{N}(i)$ — **negative set**: all k with $y_k \neq y_i$.

Supervised Contrastive Loss (SupCon) (10)

Each image receives **two independently augmented views**; the encoder produces embeddings for all $2B$ views. For anchor i , $\mathcal{P}(i)$ therefore includes both the other view of the same image *and* both views of every other same-class image in the batch.

Define the **temperature-scaled logit**:

$$\ell_{ip} = \frac{S_{ip}}{\tau}, \quad \tau = 0.07. \quad (13)$$

Small τ penalises any negative placed close to the anchor more harshly. The loss for anchor i averages over all positives, and for each one computes how it scores against the full batch:

$$\mathcal{L}_i^{\text{SC}} = -\frac{1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{e^{\ell_{ip}}}{\sum_{a \neq i} e^{\ell_{ia}}}. \quad (14)$$

The final loss is the mean over all anchors.

Geometric effect. The softmax denominator simultaneously pulls all same-class embeddings toward the anchor while pushing every different-class embedding away — an all-pairs, equal-weight contraction/expansion.

Limitation. Every positive pair contributes the same gradient regardless of whether it is easy or hard. A pair that is already well separated contributes as much as one right at the class boundary.

Multi-Similarity Loss (MS-Loss) (19)

MS-Loss addresses the uniform weighting of SupCon by up-weighting boundary-straddling pairs and discarding already-correct ones.

Step 1 — Pair mining. For anchor i , prune both sets:

$$\mathcal{P}^*(i) = \{k \in \mathcal{P}(i) \mid S_{ik} < \max_{j \in \mathcal{N}(i)} S_{ij} + \varepsilon\}, \quad (15)$$

$$\mathcal{N}^*(i) = \{k \in \mathcal{N}(i) \mid S_{ik} > \min_{j \in \mathcal{P}^*(i)} S_{ij} - \varepsilon\}, \quad (16)$$

where $\varepsilon = 0.1$ is the mining margin. This discards positives already well-separated from all negatives, and negatives already far from all positives.

Step 2 — Positive loss. For each surviving positive $k \in \mathcal{P}^*(i)$, the contribution $e^{-\alpha(S_{ik}-\lambda)}$ is large when S_{ik} is low (the pair is pulled apart) and negligible when S_{ik} is already high:

$$\mathcal{L}_i^+ = \frac{1}{\alpha} \log \left(1 + \sum_{k \in \mathcal{P}^*(i)} e^{-\alpha(S_{ik}-\lambda)} \right), \quad \alpha = 2. \quad (17)$$

Step 3 — Negative loss. Symmetrically, $e^{\beta(S_{ik}-\lambda)}$ is large when S_{ik} is high (the negative is too close):

$$\mathcal{L}_i^- = \frac{1}{\beta} \log \left(1 + \sum_{k \in \mathcal{N}^*(i)} e^{\beta(S_{ik}-\lambda)} \right), \quad \beta = 50, \quad (18)$$

where $\lambda = 0.5$ is the base cosine-similarity threshold. The asymmetry $\beta \gg \alpha$ reflects that a dangerously close negative is more harmful than a moderately far positive.

Full MS-Loss:

$$\mathcal{L}_i^{\text{MS}} = \mathcal{L}_i^+ + \mathcal{L}_i^-, \quad (19)$$

with the final loss being the mean over all anchors.

A.4.1 Comparison between SNGP and GMM

Method	NLL (mean $\pm$ std)	Accuracy (mean $\pm$ std)
SNGP	$0.3309 \pm 9.84 \times 10^{-3}$	$0.9337 \pm 1.91 \times 10^{-3}$
GMM	$1.5321 \pm 3.87 \times 10^{-2}$	$0.9286 \pm 2.40 \times 10^{-3}$

Table 5: Comparison between SNGP and GMM (Gaussian Mixture Model).

A.5 Contrastive Learning Matches Cross Entropy

Problem Setup. We consider the following modeling pipeline:

$$\mathbf{z} = h_\theta(\mathbf{x}), \quad \hat{\mathbf{y}} \sim \mathcal{GP}(\mathbf{z}), \quad p(\mathbf{y} | \mathbf{x}) = \text{softmax}(\hat{\mathbf{y}})$$

where $\mathbf{x}$ denotes an input image, $\mathbf{z}$ is the learned embedding produced by the encoder h_θ , $\hat{\mathbf{y}}$ represents the pre-softmax logits modeled via a Gaussian process, and $\mathbf{y}$ denotes the predicted class.

The supervised contrastive (SupCon) loss is computed over the embedding space $\mathbf{z}$, while the cross-entropy (CE) loss is applied to the predicted class probabilities $p(\mathbf{y} | \mathbf{x})$.

Joint Training Objective. We study the effectiveness of the supervised contrastive loss in a joint training setting, where it is combined with the standard cross-entropy objective. The overall loss function is defined as

$$\mathcal{L} = \lambda \cdot \mathcal{L}_{\text{supcon}} + \mathcal{L}_{\text{CE}},$$

where λ controls the relative contribution of the SupCon term.

We sweep λ over a wide range of values, from 0 to 256, to evaluate how strongly the contrastive signal influences representation learning and downstream performance. For each value of λ , we conduct three independent runs with different random seeds.

Findings. As shown in Fig. 8, the supervised contrastive (SupCon) loss provides a learning signal that is largely aligned with that of the cross-entropy (CE) objective. This is evidenced by the substantial overlap in error bars across different runs. In particular, varying λ over the explored range does not lead to meaningful changes in the final performance metrics. These results suggest that the two objectives encourage similar embedding structures.

Moreover, even when $\lambda = 0$, which effectively removes gradient contributions from the SupCon term, we still observe a consistent decrease in the SupCon loss during training. This behavior suggests that improvements in the learned representations induced solely by the cross-entropy objective are sufficient to reduce the SupCon loss. Overall, this provides empirical evidence that the two objectives are strongly correlated in this setting.

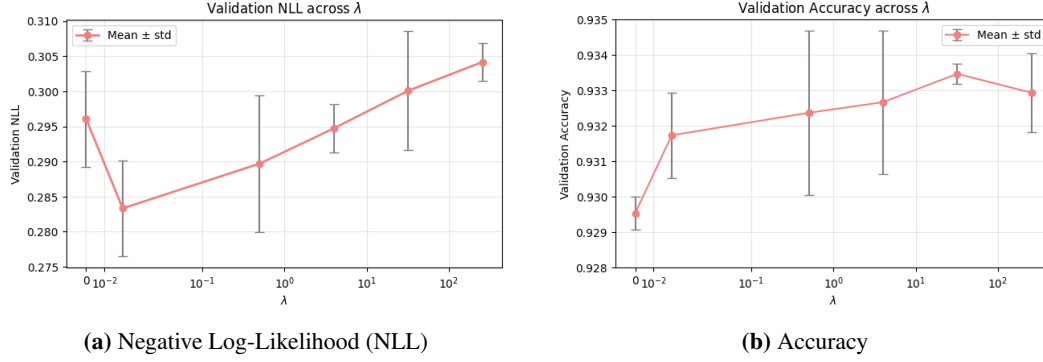


Figure 8: Effect of λ on model performance. Each point represents the mean across three runs, with error bars indicating one standard deviation.

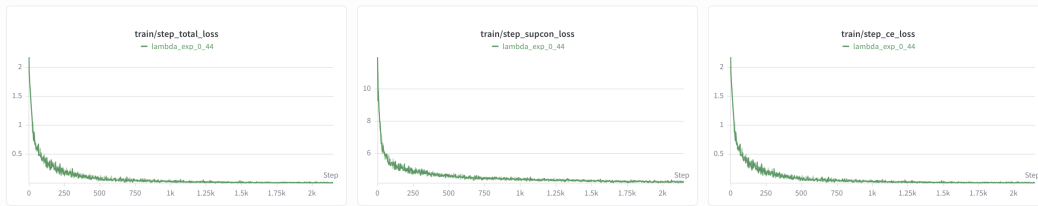


Figure 9: Training dynamics with $\lambda = 0$. The total loss is defined as $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{supcon}} + \mathcal{L}_{\text{CE}}$. As the cross-entropy loss $\mathcal{L}_{\text{CE}}$ decreases training, the supervised contrastive loss $\mathcal{L}_{\text{supcon}}$ also decreases, despite the absence of gradient propagation from the SupCon term.